Google Cloud H2S HACKATHON

AI Builder Cup

Build in JAPAC.
Build for the world.



# Team Details

a. Team name: **Techno Crackers**

b. Team leader name: **Venket Raj S** (Team: Venket Raj S · Malik Chandra Pandian · Arun Veerappan · Nagaraj — 4 working professionals in JAPAC, no students)

c. Problem Statement: **BFSI: Intelligent Risk, Fraud & Financial Experiences**

**CatSight — Event-aware contract intelligence for catastrophe reinsurance:** After a typhoon or earthquake, a reinsurer cannot say quickly or defensibly how much it owes because the answer depends on contract clauses buried in PDF treaty wordings.

# Brief about the idea

**CatSight** tells a reinsurer, **about 11 seconds** (median) after a typhoon or earthquake, which contracts pay, how much, and which clauses could change that, with every quote checked against the contract page.

**The problem, in people terms**

- After Typhoon Jebi, insurers paid **857,284 claims (¥1,067.8bn)** (GIAJ) to families and businesses.
- They recover much of it from **reinsurers**, under 50-page legal contracts called **treaties**.
- Working that out takes an estimated **30–60 analyst-hours**, and unclear clauses (wind vs. storm-surge flooding) lead to disputes. Slow payouts mean slower rebuilding.

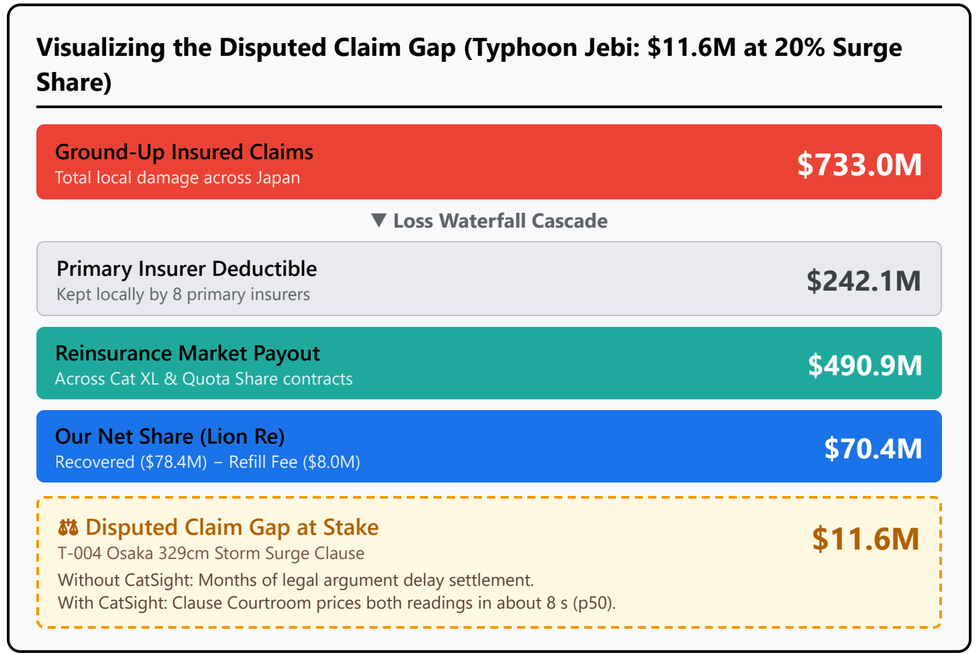
**Real disasters, fictional contracts.** Real treaties are confidential, so we replay real public hazard data over **24 synthetic treaties for 8 fictional insurers** in Japan, the Philippines and Taiwan.

| Source | Time to arrive | What it gives you |
|---|---|---|
| Catastrophe-model estimate | **4–20 days** | A market-wide range, not your contracts |
| PERILS (Asia loss index) | **6 weeks** | An industry total, not treaty payouts |
| Verisk PCS Japan | **up to 90 days** | Market index, events ≥ $2bn only |
| Manual spreadsheet review | **2–4 days** | Slow; easy to miss endorsements |
| **CatSight** | **0.7 s money · ~11 s full check** | **Your treaties, priced, with checked quotes** |

Hazard data: NOAA IBTrACS (typhoon tracks), USGS ShakeMap (earthquakes).

# Opportunities

- **How is it different?**
  - Catastrophe models estimate the loss but stop before your contracts. Document AI reads contracts but not the disaster. **CatSight joins the two, in seconds.**
  - Works **alongside** licensed models (Moody's RMS, Verisk): their footprint can plug in.
- **How does it solve the problem?**
  - **Tested code does every calculation:** money on screen in **0.7 s** (median). The AI never does maths.
  - **Gemini finds the clauses that matter;** every quote is matched word-for-word, then checked by a second model.
- **What's unique, and the social impact**
  - **Clause Courtroom:** two AI lawyers argue an unclear clause; the engine prices both readings (**$11.6M** at stake on the recorded run).
  - **Reads scanned, stamped endorsements** that have no text layer.
  - **91%** of APAC disaster losses ($65B) are uninsured; **48%** of insurers lack catastrophe models. Faster, clearer settlements help communities rebuild.



# List of features offered by the solution

| Feature | What it gives the analyst | How it stays trustworthy |
|---|---|---|
| **1. Who pays what** | Matches all 24 contracts by peril, place and dates (with a reason for each of the 11 that don't respond); payout per layer, coverage used, refill fees, and where the insurer, all reinsurers and Lion Re end up. | Tested code; exact match with a hand-worked example. |
| **2. "How was this calculated?"** | Each contract step by step: wind or shaking, damage %, insured value × damage % = loss, and each layer's formula with its citation. | Every number can be inspected. |
| **3. Clause flags & scanned pages** | Flags clauses that bite on *this* event (T-004 excludes storm surge; Jebi's 329 cm Osaka tide) and reads a scanned endorsement (168 h replaces 72 h). | Two-step quote check; failures shown as withheld; scanned pages badged *transcribed*. |
| **4. Clause Courtroom** | Two AI lawyers argue opposite readings in parallel, from checked quotes; the engine prices both; a neutral arbiter summarises. | Issue-spotting only; the analyst records the decision. |
| **5. What-if & follow-up questions** | Try a labelled hypothetical storm window; ask plain questions about the contracts and results. | Bounded inputs; questions screened by a local filter + **Google Cloud Model Armor**. |
| **6. Replay, tour & reports** | 6-second free replay of a real run, 9-stop guided tour, colour-blind-safe themes, Word & Markdown reports. | Every report carries the method version, check counts and disclaimers. |

# Process flow diagram or Use-case diagram

### Workflow 1: live analysis (ADK 2.11 `Workflow`)

**1. event_intel**
Loads the real storm track or earthquake shaking map, and the event's timeline

▼

**2. exposure_analyst (no AI: tested Python)**
Matches 24 treaties → loss by region → payout per layer. Money on screen in ~0.7 s

▼

**3. wording_checker (Gemini 3.5 Flash-Lite)**
Searches each treaty's clauses (Firestore vector search) → candidate flags with page, clause and quote

▼

**4. verifier ⇄ reviser (one retry)**
Word-for-word quote match, then Gemini 3.8 Flash checks the quote supports the claim → one retry → else withheld

▼

**5. reporter (Gemini 3.5 Flash-Lite + number check)**
Writes the memo from verified facts; code checks every number against the engine

### Workflow 2: Clause Courtroom (parallel)

**1. brief**
Picks the most material verified exclusion flag and the clauses around it

▼

**2a. cedent_counsel**
Flash-Lite · argues the insurer's reading + excluded share (0–50%)

**2b. reinsurer_counsel**
Flash-Lite · argues the reinsurers' reading + excluded share (0–50%)

**▼ Join + quote & support check (one retry per side)**

**3. price (tested Python)**
Prices both readings through the treaty layers; money at stake for each party

▼

**4. arbiter (Gemini 3.8 Flash) → the analyst decides**
Summarises both sides without deciding; the analyst records a position

# Wireframes/Mock diagrams of the proposed solution (optional)

### Analyst Cockpit: 5-Step Progressive UI Wireframe

React 18 + SSE Stream

https://catsight-api-342162443546.us-central1.run.app

CatSight · Typhoon Jebi | Local Claims: $733.0M | Reinsurers: $490.9M | Our Net: $70.4M | ✓ 1 Verified · 1 Withheld

1. Event Track | 2. Who's Hit | 3. What It Costs | 4. Clauses | 5. Decide

**Exposure Choropleth Map**




34 prefectures overlaid on peak gust contours

**Clause Courtroom & Layer Stack**

Insurer Counsel
"Windstorm caused collapse; surge excluded."
**0% Surge share**

Reinsurer Counsel
"Flooded by storm tide; clause 8 bites."
**20% Surge share**

Disputed Gap: $11.64M priced live (at 20% surge share)

Instant Replay: **6 s** · Live Agent SSE: **11.3 s p50** | Python Decimal: **Zero LLM arithmetic**

### Trust Architecture: 2-Stage Verification Funnel

Every quote checked

**1. Candidate AI Clause Flag / Counsel Argument**
Model proposes: `treaty_id`, `page`, `clause_no`, and verbatim `quote`

▼ Filter 1

**Stage 1: Deterministic Word-Aligned Quote Matcher** — 0 ms Python
Normalised substring + ellipsis ( `...` ) matcher against Firestore treaty PDF text chunk. Catches 40/40 corrupted quotes; tags scanned endorsements as `transcribed`.

▼ Filter 2

**Stage 2: Semantic Support Check (Gemini 3.8 Flash)** — ~1.2 s Reasoning
Verifies quote + event timeline ( `start`, `end`, `duration_h` ) + JMA facts directly support the exclusion. 1 revision retry loop on failure.

**✓ Verified Citation**
Rendered in UI with clickable source PDF page pill & pricing impact

**⊘ Visibly Withheld**
Suppressed from findings; shown as `n verified · k withheld`

# Architecture diagram of the proposed solution

**Analyst / judge browser (React 18 + TypeScript)**
One origin over HTTPS · results stream live (server-sent events)

▼

**Google Cloud Run — `catsight-api` (us-central1)**
One container: UI + FastAPI · ADK 2.11 `Workflow` graphs · tested Python loss engine · least-privilege service account · max 3 instances · hourly & daily caps

▼

**Gemini Enterprise Agent Platform**
(formerly Vertex AI)
- 3.5 Flash-Lite
- 3.8 Flash
- embedding-001

**Firestore**
- Treaty book & events
- 252 clause vectors
- Saved runs

**Cloud Storage**
- Treaty PDFs
- Stamped endorsements

**Model Armor**
- Screens follow-up questions & answers

▼

**Cloud Logging · Trace · Monitoring · Log Analytics**
A log line per stage · guardrail metrics (citations withheld, lawyers dropped, unsupported figures) · live dashboard

**Key decisions**

- **Money first, AI second:** tested code prices the loss before any AI call (0.7 s). The AI never calculates.
- **Rules are data, not code:** treaty terms, damage curves and market rules live in versioned data files; the version in force is stamped on every run.
- **One service, no keys:** UI and API on one Cloud Run service; a dedicated service account, no API keys.
- **Ready for judging:** pre-computed searches (no embedding calls on a cold start), one warm instance, and rate limits that fall back to the recorded replay.

# Technologies to be used in the solution

| Technology / Service | Role | Why we chose it |
|---|---|---|
| **Gemini 3.5 Flash-Lite**<br>Gemini Enterprise Agent Platform | Agents & page reading | Fast, low-cost structured output for wording checks, the two Courtroom lawyers, report text and the scanned page. |
| **Gemini 3.8 Flash**<br>Gemini Enterprise Agent Platform | Checking & arbiter | Stronger reasoning to check that each quote supports its claim, and to summarise the Courtroom fairly. |
| **Google ADK 2.11** | Agent workflows | Graph `Workflow`: a 5-step analysis with a bounded retry, and a Courtroom with two lawyers in parallel ( `JoinNode` ). |
| **Firestore + Vector Search**<br>`gemini-embedding-001` | Data & clause search | Treaty book, 252 clause vectors filtered by treaty (186/186 governing clauses found), runs and audit trail. |
| **Google Cloud Model Armor** | Question screening | Screens follow-up questions and answers for prompt injection, jailbreaks and harmful content, after a local filter. |
| **Cloud Run & Cloud Storage** | Hosting & PDFs | One container for UI + API; scale-to-zero by default, one warm instance during judging; Storage holds the treaty PDFs. |
| **Cloud Logging, Trace & Monitoring** | Observability | A log line per stage, per-agent traces, and a live dashboard built from log-based metrics. |
| **Python 3.12 · FastAPI · React 18** | Engine & UI | Tested loss engine (870+ offline + 13 live tests), Word report generator, accessible TypeScript UI. |

# Estimated implementation cost (optional)

**$0.010**
Median AI cost per live analysis ($0.008–$0.014)

**$0.005**
Median AI cost per Clause Courtroom run ($0.005–$0.006)

**$0.000**
Instant Replay (a recorded real run, no AI calls)

**$0.003**
Follow-up question (from token counts)

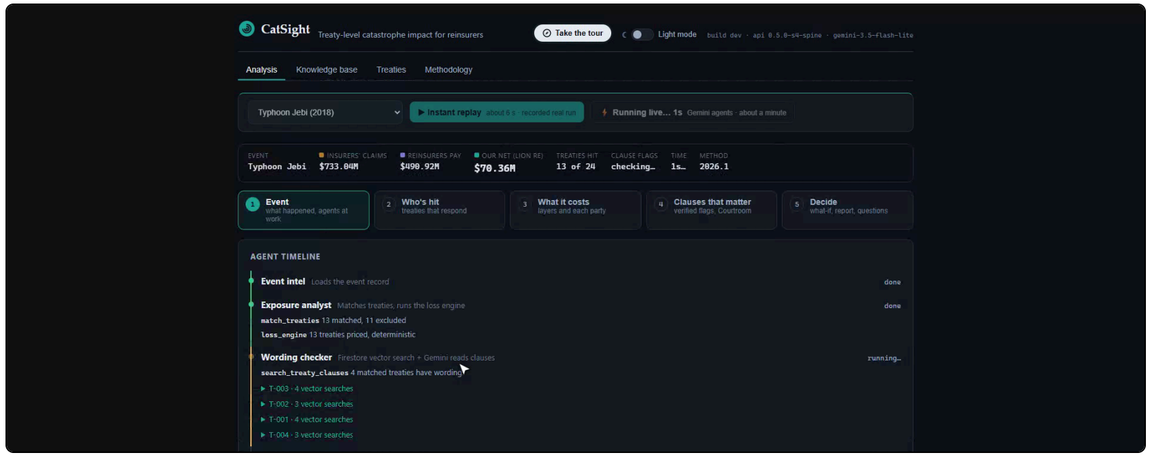
### Measured Cloud & Token Breakdown

| Workload | Model / Service | Measured Cost |
|---|---|---|
| One-time ingestion (8 PDF wordings, 252 chunks) | `gemini-embedding-001` + `3.5-flash-lite` | < **$0.08 total** (57 s runtime) |
| Live event analysis (13 responding treaties) | `3.5-flash-lite` + `3.8-flash` verifier | **$0.0097 p50** ($0.0081–$0.0140) |
| Clause Courtroom (parallel counsel + arbiter) | 2× `3.5-flash-lite` + `3.8-flash` | **$0.0052 p50** ($0.0047–$0.0064) |
| Cloud Run + Firestore | `min-instances 0` by default (1 during judging), `max 3` | **~$0 idle** at zero instances · pennies per active hour |

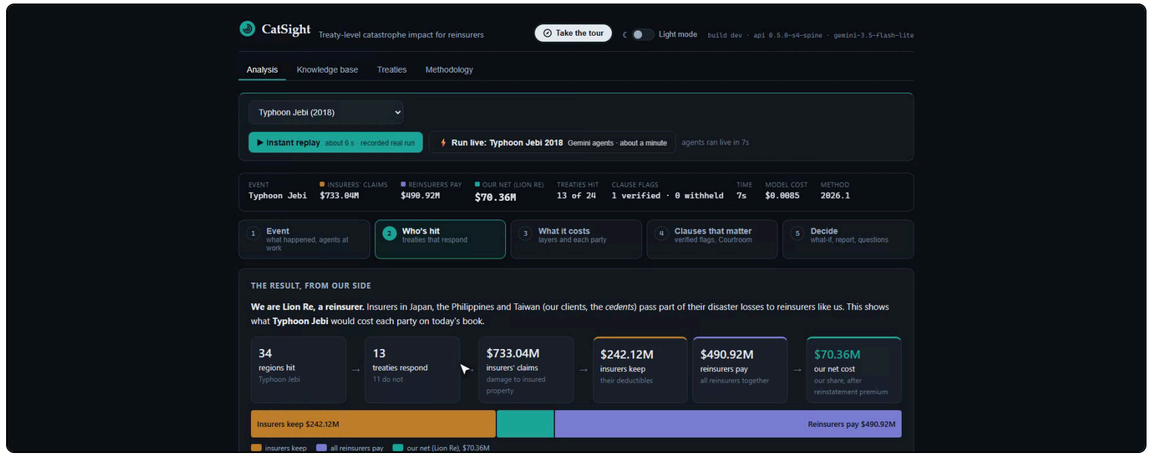
### Annual Run-Rate vs. Manual Analyst Effort

- **A mid-size reinsurer:** 8 major events + 20 smaller checks a year (≈ 500 runs) → **~$7/year of Gemini cost**, plus ~$25–$40/month for Cloud Run and Firestore when active.
- **Analyst time saved (estimate):** 30–60 hours per event down to 3.5–8 hours → **165–675 hours a year** (about $12k–$100k; labelled *simulated*).
- **Cost guards:** 20 live runs/hour per visitor, 300/day cap, automatic fallback to Instant Replay, billing budget alerts.

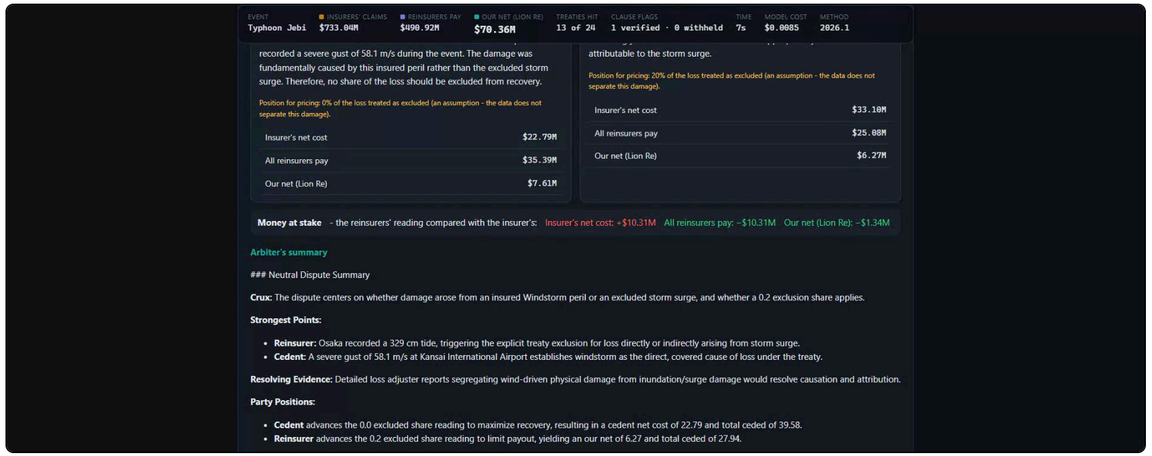
# Snapshots of the prototype



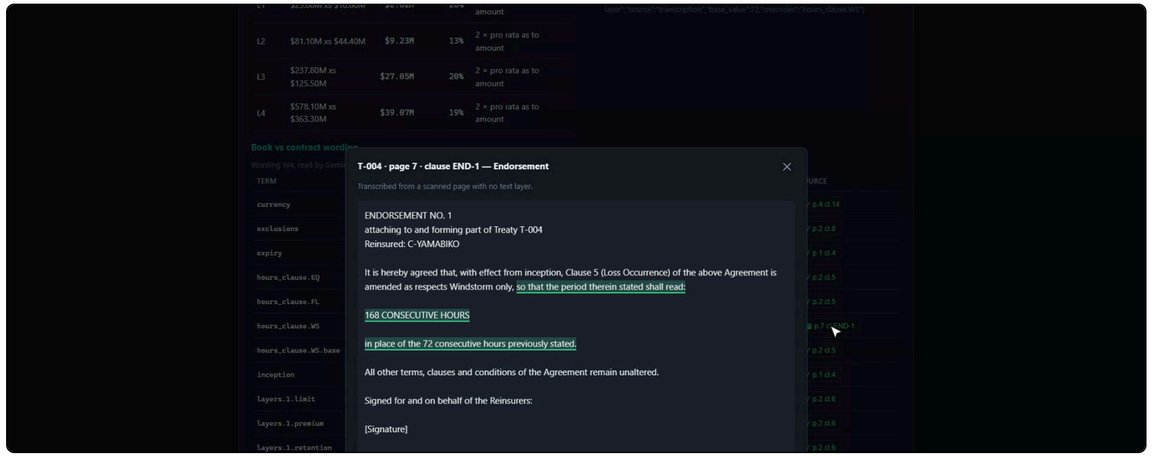

**1. Live Agent Streaming:** ADK 2.11 stages stream over SSE; deterministic loss figures arrive in ~0.7 s while Gemini retrieves & verifies treaty clauses.




**2. Three-Party Loss Waterfall:** 13 of 24 treaties respond on Typhoon Jebi ($733.04M claims → $242.12M insurer retention, $490.92M reinsurers, $70.36M Lion Re net).




**3. Clause Courtroom (Jebi T-004 Storm-Surge Exclusion):** Parallel counsel argue 0% vs 20% surge share; engine prices both (Ceded −$11.64M, Insurer +$10.31M, Reinsurers −$10.31M, Lion Re −$1.34M).




**4. Scanned Endorsement Detective (T-004 p.7):** Gemini transcribes a rasterised stamped endorsement with no text layer (168 hours replaces 72 hours) and badges it `transcribed`.

# Prototype Performance report/Benchmarking

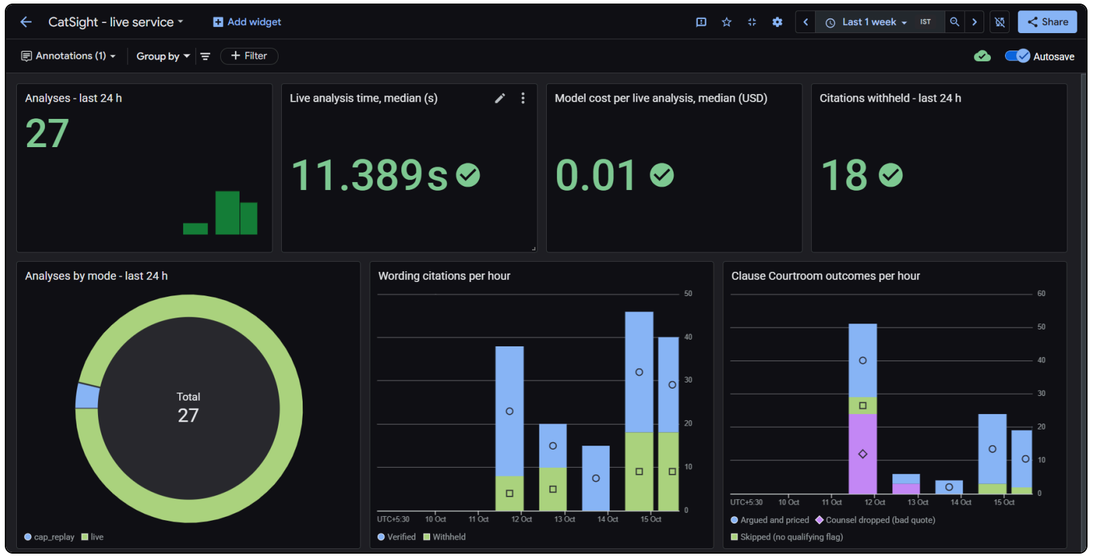
**0.7 s**
Money on screen (median; 0.5–1.1 s over 10 live runs)

**11.3 s**
Full analysis with checked quotes (median; 7.8–19.3 s)

**99.5%**
Treaty terms extracted (185/186 fields, in 57 s vs ~25 min)

**870+**
Automated tests (870+ offline + 13 live; CI on GitHub)

| What we measured ( `docs/RESULTS.md` ) | Result |
|---|---|
| **Loss engine vs. hand-worked example**<br>every layer, reinstatement and quota-share formula | **100% exact** |
| **Clause search & citation audit**<br>1 scanned page transcribed; 1 unverified withheld | **186/186 found · 185/186 verified** |
| **Clause Courtroom, live**<br>skipped in 2 when the T-004 flag was withheld; $10.3M–$26.0M at stake priced live | **Ran in 8 of 10; both sides verified in all 8** |
| **Bad-quote rejection** (offline test suite) | **40/40 rejected, 0 false withholds** |
| **Untraceable numbers in AI-written text** | **0 across all live runs** |




The live-service dashboard in Cloud Monitoring (screenshot, 15 Oct 2026). "Citations withheld" counts quotes the checker refused to show: the guardrail working, not an error.

**Honest limits:** extraction prompts were tuned on the same 8 synthetic wordings (99.5% is an upper bound; a blind set is next). Analyst-time savings are labelled *simulated estimates*.

# Additional Details/Future Development (if any)

### Why pricing each party matters

**1. Jebi Courtroom (T-004 storm-surge exclusion, p.2 cl.8):**
At a 20% surge share (the recorded run), reinsurance payouts fall **−$11.64M** ($39.58M → $27.94M); net of refill fees, reinsurers save **$10.31M** and the insurer carries **+$10.31M**; Lion Re saves **$1.34M**. The demo video shows a separate 30% run: **−$17.46M** ceded, Lion Re **−$2.44M**.

**2. Hours clause (worked example, What-if):**
Splitting one $54.4M event into two ($35.0M + $19.4M) cuts reinsurers' recoveries by **−22.5% (−$10.0M)**, yet **raises Lion Re's loss +7.4%** ($6.82M → $7.33M), because the split moves loss into the layer where Lion Re's share is largest.

### Team Techno Crackers — Roles

• **Venket Raj S (Lead):** Architecture, Cloud Run / Firestore / IAM deployment & observability
• **Malik Chandra Pandian:** Reinsurance domain modelling, `Decimal` loss engine & worked examples
• **Arun Veerappan:** ADK 2.11 workflows, RAG extraction, citation verifier & Clause Courtroom
• **Nagaraj:** React 18 + TypeScript + custom SVG choropleth UI, 5-step analyst workflow, tour & exports

### Future Development Roadmap (Designed / Investigated)

- **A real hours-clause dispute:** the 2024 Hualien earthquake (Taiwan). Its real aftershocks ~20 days later are a second occurrence worth ~$11–15M per treaty.
- **Damage curves per country:** calibrated for the Philippines, bringing Typhoon Haiyan back into the demo.
- **Licensed hazard footprints:** Moody's RMS or Verisk behind our hazard interface.
- **Your own contracts:** authenticated treaty upload, and BigQuery for portfolio-scale exposure.
- **Automatic start:** USGS / GDACS disaster alerts via Pub/Sub, plus an A2A/MCP endpoint for other agents.

# Provide links to your:

- **GitHub Public Repository**
  https://github.com/venketraj/catsight-ai-buildercup-2026
  Apache-2.0 licence · 870+ offline tests + 13 opt-in live tests · GitHub Actions CI · Full architecture & assumptions in `docs/`.
- **Demo Video Link (under 3 minutes: 2 min 45 s)**
  https://drive.google.com/file/d/14Yv-A-KtVksOgRxOxafZcEKFBUvFumQi/view?usp=sharing
  The problem, a live Typhoon Jebi analysis, a scanned endorsement read by Gemini, and the Clause Courtroom.
- **Final Product Link**
  https://catsight-api-342162443546.us-central1.run.app
  Live on Google Cloud Run · No login · **Instant replay** (~6 s) or **Run live** (~11 s median) · 9-stop tour · 3 example questions built in · the README has a 2-minute judge guide.
- **Submission Folder (Google Drive)**
  https://drive.google.com/drive/folders/1X5cNQh4aMP-B1JIvIXyyxCL-4anxJhl4?usp=sharing
  The demo video and this deck (PDF).

**Disclaimers & Data Attribution:** Indicative first view on synthetic data. Not a catastrophe-model output. Not underwriting or claims advice. Issue-spotting only — not legal advice. Jebi, Hagibis, Haiyan and Noto were real events in which people died. Portfolio, insurers and treaty wordings are 100% synthetic; hazard data is public (NOAA IBTrACS v04r01, USGS ShakeMap, Natural Earth).

Google Cloud H2S

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# Thank you

